

(192565006)

38

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I Thanigaivel S. (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Thanigaivel.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with	Clear explanation	Limited explanation; lacks depth	Poor or unclear explanation

20/10

Re

		relevant examples	with adequate details		
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well- structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandabl e with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

2. Sliding window.

- Window Size = 6.
- Total frames = 48.
- Windows required = $48 \div 6 = 8$.
- One frame lost per window.
- Retransmissions = 8.

Explanation: Flow control prevents the sender from overwhelming the receiver, improving efficiency and reducing packet loss.

3. IP Fragmentation:

- Packet Size = 12,000 bytes.
- MTU = 1500 bytes.
- IP header = 20 bytes.
- Data per fragment = $1500 - 20 = 1480$ bytes.
- Fragments = $\text{Ceil}(12000 \div 1480) = 9$.
- Header overhead = $9 \times 20 = 180$ bytes.

Explanation: The network layer fragments packets, assigns fragments offsets and reassembles them at the destination.

4. Given:

- Window size = 6 frames.
- Total frames = 48.

$$\text{Windows} = \frac{48}{6} = 8.$$

$$\text{Retransmission} = 8 \times 1 = 8.$$

Explanation: Flow control matches the sender's transmission rate to the receiver's capacity, ensuring efficient communication.

5. Session later.

- File = 600MB
- Checkpoint every 60MB
- failure after 390MB
- checkpoints before failure: 60, 120, 180, 240, 300, 360, = 6 checkpoints
- Retransmission = 390 - 360 = 30MB

Explanation: Synchronization checkpoints allow communication to resume from the latest checkpoint instead of restarting.